State-Vector Rap Sheet 2

Name: Singlet State (Maximum Entanglement)

Wanted for: Nonlocality, Complete Quantum Weirdness

Description: The composite system as a whole is fully characterized. There is no information about Alice's or Bob's subsystems.

State-Vector: $\frac{1}{\sqrt{2}}\left(|ud\rangle - |du\rangle\right)$

Normalization: $\psi^*_{uu}\psi_{uu} + \psi^*_{ud}\psi_{ud} + \psi^*_{du}\psi_{du} + \psi^*_{dd}\psi_{dd} = 1$

Density Matrix:
Full Composite System: $\rho^2 = \rho$, and $Tr(\rho^2) = 1$.
Alice's Subsystem: Density matrix is proportional to the unit matrix, having equal eigenvalues that add up to 1. Hence, each measurement outcome is equally likely. $\rho^2 \neq \rho$, and = $Tr(\rho^2) < 1$.

Wave Function: Not Factorized: $\psi(a, b)$

Expectation Values:

$$\langle\sigma_z\rangle, \langle\sigma_x\rangle, \langle\sigma_y\rangle = 0$$
$$\langle\tau_z\rangle, \langle\tau_x\rangle, \langle\tau_y\rangle = 0$$
$$\langle\tau_z\sigma_z\rangle, \langle\tau_x\sigma_x\rangle, \langle\tau_y\sigma_y\rangle = -1$$

Correlation: $\langle\sigma_z\tau_z\rangle - \langle\sigma_z\rangle\langle\tau_z\rangle = -1$

State-Vector Rap Sheet 3

Name: "Near-Singlet" (Partial Entanglement)